

June-Seop Yoon

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High-Performance HCI Lab

RESEARCH INTERESTS

My research interests lie in computational modeling of esports players' behavior and its applications to player behavior analysis, performance evaluation, and personalized training in the gaming and esports context. I focus primarily on first-person shooter (FPS) games, motivated by twenty years of experience as an avid FPS player and by the significant potential of computational approaches for advancing scientific behavior analysis in this genre. I am currently exploring model-based player training, an emerging approach that leverages computational behavior models to design personalized training protocols and provide optimal feedback for players.

EDUCATION

Yonsei University

Seoul, Korea

Ph.D. candidate in Computer Science, GPA: 4.34/4.50, Advisor: Prof. Byungjoo Lee August 2021–August 2026

- Thesis: “Computationally Rational Aim-and-Shoot Behavior Model in First-Person Shooters: Player-State Inference and Personalized Training Design”

Yonsei University

Seoul, Korea

B.S. in Computer Science, GPA: 4.03/4.50 March 2016–August 2021

PUBLICATIONS AND MANUSCRIPTS

- **June-Seop Yoon**, Hee-Seung Moon, Ben Boudaoud, Josef Spjut, Iuri Frosio, Byungjoo Lee, Joohwan Kim, “Modeling visually-guided aim-and-shoot behavior in first-person shooters”, *International Journal of Human-Computer Studies*, 2025. [DOI](#)
- Jonghyun Kim, Joongseok Kim, **June-Seop Yoon**, Hee-Seung Moon, Sunjun Kim, Byungjoo Lee, “Modeling User Performance in Multi-Lane Moving-Target Acquisition”, *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 1–18, 2025. [DOI](#)
- Donghyeon Kang, Namsub Kim, Daekaun Kang, **June-Seop Yoon**, Sunjun Kim, Byungjoo Lee, “Quantifying wrist-aiming habits with a dual-sensor mouse: Implications for player performance and workload”, *CHI 2024*. [DOI](#)
- Jimoon Kang[†], **June-Seop Yoon**[†], Byungjoo Lee, “How AI-based Training Changed Performance Development of Go Players”, *CHI 2022*. [DOI](#)
- Yong-Hun Cho, Dae-Hong Min, Jin-Suk Huh, Se-Hee Lee, **June-Seop Yoon**, In-Kwon Lee, “Walking Outside the Box: Estimation of Detection Thresholds for Non-Forward Steps”, *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, 448–454, 2021. [DOI](#)
- **June-Seop Yoon**[†], Soomin Kim[†], Hanbyeol Lee, Michael Xuelin Huang, David Lindlbauer, Shumin Zhai, Hee-Seung Moon, Byungjoo Lee, “Hierarchical Amortized Inference for Simulation Models in HCI”, manuscript under review

INVITED TALKS

- NVIDIA Graphics Forum, “Modeling Computationally Rational Aim-and-Shoot Behavior in First-Person Shooters”, August 20, 2025

RESEARCH AND INDUSTRY EXPERIENCE

YESLAB Inc.

Technical Contributor / Shareholder

Korea

2024

- Provided an FPS player skill-evaluation solution for T1 Academy

Aalto University

Visiting Student, Advisor: Prof. Antti Oulasvirta

Finland

December 2022–January 2023

NVIDIA Research

Research Intern in HPX team, Mentor: Dr. Joohwan Kim

Remote

June 2022–September 2022

AWARDS & SCHOLARSHIPS

- Encouragement Award, Graduate Innovation Excellent Paper Competition, Yonsei University (Spring 2026)
- Excellence Award, 1st Yonsei AI Workshop Paper Contest, Yonsei University (2022)
- Full Scholarship for Academic Excellence, Yonsei University (Fall 2021–Spring 2023)

TEACHING EXPERIENCE

- **Human and Computer Interface (CSI4107)**: Teaching Assistant, Fall 2021, Fall 2022
- **Computational Interaction Design (CSI6545)**: Teaching Assistant, Fall 2021
- **Computer Programming (CAC1100)**: Teaching Assistant, Spring 2022
- **Artificial Intelligence Programming (AIC2100)**: Teaching Assistant, Spring 2023, Spring 2024
- **Esports Performance Engineering (CSI8200)**: Teaching Assistant, Spring 2023

ACADEMIC SERVICE

- **Reviewer**: CHI, IJHCS, CHI PLAY, JEGE

SKILLS

- **Python, C++, Matlab**
- **NumPy, Pandas, SciPy, PyTorch, Stable-Baselines3, Panda3D, Godot**
- **Computational behavior modeling, Computational interaction design, Reinforcement learning, Machine learning, User study design**
- **Korean** (Native), **English** (Fluent), **Japanese** (Intermediate)

GAMING EXPERIENCE

- **Team Fortress 2**: 3000+ hours since 2012
- **Overwatch**: 5000+ hours since 2016; 4200+ rating in 2018; Cyber Yonsei-Korea Games champion in 2019